



SEMESTER 1 EXAMINATIONS 2024/2025

MODULE: CSC1110 - Natural Language Technologies

PROGRAMME(S):

DS	BSc Data Science
ECGTE	Engineering & Computing GTE

YEAR OF STUDY: 1,4

EXAMINER(S):

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TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer Question 5 and three other questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1

[TOTAL MARKS: 25]

Q 1(a)

[6 Marks]

Consider the following six features used to train a sentiment analysis system:

f1: how many explanation marks are in the document?

f2: how many pronouns are in the document?

f3: how many words in the document appear on the positive side of a sentiment lexicon?

f4: how many words in the document appear on the negative side of a sentiment lexicon?

f5: how many words containing all capitalized letters, e.g. NOT, are in the document?

f6: how many times does the word 'not' occur in the document?

During training the following weights are learned:

w1: 0.11

w2: 2.4

w3: 3.5

w4: -3.7

w5: -1.2

w6: -0.5

bias: 0.2

Step through how a Logistic Regression classifier would classify the following document:

WHAT an amazing film!!! Everything from the acting to the cinematography was superb. You will not be disappointed!

Note that the words *superb* and *amazing* occurs on the positive side of a sentiment lexicon and the word *disappointed* occurs on the negative side.

Q 1(b)

[6 Marks]

The following are word probabilities estimated from a training corpus with 1000 positive reviews and 1000 negative reviews:

$p(\text{will} \mid \text{pos}) = 0.01$

$p(\text{will} \mid \text{neg}) = 0.015$

$p(\text{not} \mid \text{pos}) = 0.02$

$p(\text{not} \mid \text{neg}) = 0.03$

$p(\text{you} \mid \text{pos}) = 0.02$

$p(\text{you} \mid \text{neg}) = 0.01$

$p(\text{be} \mid \text{pos}) = 0.15$

$p(\text{be} \mid \text{neg}) = 0.15$

Step through how a Naïve Bayes sentiment classifier with words as features would classify the review:

you will not be disappointed

Note that the word *disappointed* does not occur in the training data.

Q 1(c)

[6 Marks]

Outline the relative advantages and disadvantages of the two approaches to sentiment analysis shown in Q1(a) and Q1(b).

Q 1(d)

[7 Marks]

Explain how the weights in a Logistic Regression sentiment classifier, such as the one in Q1(a), are learned.

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[6 Marks]

Write out the probability of a word sequence, $p(w_1 \dots w_n)$, according to

- a) a unigram language model
- b) a trigram language model
- c) a 5-gram language model

Q 2(b)

[6 Marks]

Provide one advantage and one disadvantage of training a 4-gram language model compared to a 3-gram language model. Explain your reasoning.

Q 2(c)

[6 Marks]

Describe how language models are evaluated.

Q 2(d)

[7 Marks]

Explain why neural language models are widely considered to be superior to n-gram language models. Explain why transformer neural language models are widely considered to be superior to the earlier feedforward neural language models. Use one or more examples in your explanation.

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[6 Marks]

Given a sequence of n words and a tagset of k tags, how many possible ways are there to tag the sequence? Explain how dynamic programming provides an efficient way to find the most likely tag sequence.

Q 3(b)

[6 Marks]

Provide three examples of feature templates that may be used by a linear chain Conditional Random Fields sequence labeller. Explain how feature templates are mapped into actual features.

Q 3(c)

[6 Marks]

What is the difference between an emission probability and a transition probability in a Hidden Markov Model sequence labeller? Explain how these probabilities are estimated from a training corpus.

Q 3(d)

[7 Marks]

Explain, using an example, how a Recurrent Neural Network is suited to the task of sequence labelling.

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 25]

Q 4(a)

[5 Marks]

Explain, using an example, how and why TF-IDF is used to weight word counts in document vectors.

Q 4(b)

[5 Marks]

Explain, using an example, why cosine similarity is often used instead of dot product to measure the similarity between two vectors.

Q 4(c)

[5 Marks]

What is meant by the term *hidden vector*? Explain how a hidden vector is produced.

Q 4(d)

[5 Marks]

Given the following text excerpt

...melt the **chocolate** over a low heat...

explain how positive and negative training instances would be generated for the target word **chocolate** using word2vec (skipgram negative sampling) with a context window of ± 2 words.

Q 4(e)

[5 Marks]

Explain, using an example, the difference between word vectors that are produced by neural networks such as BERT and word vectors that are produced by techniques such as word2vec or GloVe.

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Q 5(a)

[5 Marks]

Briefly discuss the ethical implications of *training* large language models.

Q 5(b)

[12 Marks]

Describe what you consider to be three *ethical* and three *unethical* applications of large language models. Justify your choices.

Q 5(c)

[8 Marks]

Describe what you think will be the

- a) short-term (< 5 years)
- b) long-term (>5 years)

impact on society of large language models.

[End of Question 5]

[END OF EXAM]